

Same Prompt, Different Answer: Hidden Non-Determinism in LLM APIs Undermines Scientific Reproducibility

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Abstract

Thousands of published studies across medicine, climate science, and engineering report results generated by large language model APIs under settings documented as deterministic, yet we show that this guarantee does not hold in practice. In 4,104 controlled experiments across eight models and five API providers, API-served models reproduce their own outputs only 22% of the time under temperature-zero greedy decoding with fixed seeds, while locally deployed models achieve 96%—a gap exceeding four-fold. This variation is invisible to researchers: outputs remain semantically equivalent (BERTScore F1 > 0.97) while differing textually, meaning that non-identical results escape detection without systematic comparison. The problem amplifies in multi-turn and retrieval-augmented generation workflows, where one model produces zero exact matches across 50 runs. A quasi-isolation experiment attributes the cause to production infrastructure complexity rather than cloud deployment itself, showing the problem is solvable. We provide a lightweight provenance protocol (<1% overhead) grounded in W3C PROV that makes this hidden variation detectable and auditable, offering practical infrastructure for reproducible LLM-based research.

Keywords: Reproducibility, Large language models, Non-determinism, Provenance, API inference

Large language models (LLMs) are rapidly becoming standard research instruments. They encode clinical knowledge[1], assist diagnostic workflows[2], accelerate literature review and data extraction[26], and are increasingly integrated into peer review at scale[3]. A growing body of work deploys LLMs as evaluators of human communication[13], as psychometric assessment tools[14], and as components of multi-agent scientific pipelines[4]. When researchers use API-based models, they trust that identical configurations will yield identical outputs. API documentation reinforces this trust by offering a temperature parameter set to zero for “deterministic” behavior, a setting that should collapse the output probability distribution to a single token at each step. Our experiments show that this guarantee does not hold.

The broader reproducibility crisis in science is well documented: over 70% of researchers have failed to reproduce another scientist’s experiment[21], and the problem is acute in AI, where only 6% of papers provide sufficient information for reproduction[23]. The field faces its own “reproducibility crisis”[22], with growing concerns that AI may worsen the problem across disciplines[24]. Data leakage alone has been documented across 17 machine learning subfields[25], and leading AI journals have begun adopting structured reproducibility mechanisms in response[19]. Yet despite this attention to training-time reproducibility, the reproducibility of *inference*, the actual deployment of models to generate outputs, remains largely unexamined. This gap matters because inference is the stage at which LLMs interact with research data and produce the outputs that enter scientific records, policy documents, and clinical workflows. Existing experiment-tracking tools such as MLflow[50] were designed for training pipelines and numerical metrics, not for inference-time text-generation provenance.

We conducted 4,104 controlled experiments across eight models and five independent API providers and found that even under temperature = 0 with fixed random seeds, API-served models reproduce their own outputs only 22.1% of the time, while locally deployed open-weight models achieve 95.6% exact reproducibility, a gap exceeding four-fold (95% bootstrap CI for the ratio: 2.48–3.61×). This non-determinism is invisible to users: outputs are semantically equivalent (BERTScore F1 > 0.97) but textually different, meaning researchers cannot detect the variation by reading outputs. The phenomenon is not confined to single queries. As LLMs move into agentic pipelines that design experiments and generate hypotheses[4], this hidden variation compounds at every step. Recent editorials have underscored the urgency of transparency in multi-agent AI systems[5], yet the baseline reproducibility of the individual API calls has never been systematically quantified.

Here, we make four contributions. First, we reveal and quantify hidden non-determinism across all five major API providers tested (OpenAI, Anthropic, Google, DeepSeek, Perplexity), showing that it is a general property of cloud-served LLM APIs rather than a provider-specific anomaly. Second, we show that this non-determinism

persists and amplifies in multi-turn and retrieval-augmented generation (RAG) workflows, where Claude Sonnet 4.5 produces zero exact matches across 50 RAG runs. Third, we demonstrate that cloud deployment *per se* does not cause non-determinism: the same LLaMA 3 8B architecture served via Together AI’s cloud endpoint achieves near-local exact match rates, implicating production-infrastructure complexity (tensor parallelism, speculative decoding, dynamic batching) rather than the cloud medium. Fourth, we provide a lightweight provenance protocol grounded in W3C PROV[6], extending Model Cards[17] and Datasheets[18] to the inference layer, that adds less than 1% overhead and makes this invisible variation visible, auditable, and attributable.

Results

API-based models fail to reproduce outputs under deterministic settings

We evaluated eight models across two core tasks, structured extraction (JSON output) and scientific summarization (free text), under greedy decoding (temperature = 0) with fixed random seeds, the configuration that API documentation presents as deterministic. The results reveal a stark divide (Table 1, Fig. 1).

Locally deployed models achieve near-perfect to perfect bitwise reproducibility. Gemma 2 9B produces exact match rate (EMR) = 1.000 [1.00, 1.00] across all tasks and conditions: every output is character-for-character identical across five repetitions, confirmed by SHA-256 hash comparison. LLaMA 3 8B attains EMR = 0.987 [0.96, 1.00] for structured extraction and 0.947 [0.89, 0.99] for summarization; Mistral 7B achieves 0.960 [0.88, 1.00] and 0.840 [0.72, 0.96], respectively. The minor deviations in LLaMA 3 and Mistral are attributable to a cold-start effect: post-hoc analysis reveals that in all seven non-unanimous abstract groups, the first repetition was the sole outlier, likely due to GPU cache state initialization on the first forward pass. Discarding a single warm-up inference yields EMR = 1.000 for both models (Extended Data Table 5).

API-served models tell a starkly different story. Under the same greedy decoding with fixed seeds, GPT-4 achieves EMR = 0.443 [0.32, 0.57] for extraction and 0.230 [0.16, 0.30] for summarization. Claude Sonnet 4.5 achieves 0.190 [0.05, 0.40] and 0.020 [0.00, 0.05], respectively; across 50 pairwise comparisons, effectively no two summarization outputs were character-identical. Perplexity Sonar falls further to EMR = 0.100 for extraction and 0.010 for summarization. The overall local average EMR is 0.956 versus 0.221 for API models, meaning that fewer than one in four API output pairs are identical under conditions documented as deterministic. This gap survives Holm–Bonferroni correction across 68 hypothesis tests (51 significant at $\alpha = 0.05$; Supplementary Section S9) and is confirmed by large Cliff’s delta effect sizes ($\delta = 0.784\text{--}0.896$), paired *t*-test (Cohen’s $d > 1.6$), and a balanced 10-abstract subsample analysis that controls for sample-size differences between models (local EMR = 0.953 vs. API EMR = 0.304, $3.1\times$ gap; Extended Data Table 4). A chat-format control experiment (200 runs) confirmed that the prompt format (completion versus chat API) does not contribute to the observed variation (Supplementary Section S7).

Non-determinism varies substantially across providers

The five API providers span a wide reproducibility range, with an 80-fold difference between the most and least reproducible (Fig. 1). DeepSeek Chat achieves the highest API reproducibility (EMR = 0.800 for extraction, 0.760 for summarization), approaching local-model performance. GPT-4 occupies an intermediate position (EMR = 0.443 for extraction, 0.230 for summarization), with the extraction–summarization gap suggesting that task output structure mediates reproducibility, as JSON-constrained outputs leave less room for lexical variation than free-text summaries. Claude Sonnet 4.5 (0.190/0.020) and Perplexity Sonar (0.100/0.010) occupy the low end, while Gemini 2.5 Pro (evaluated on multi-turn and RAG tasks only) achieves EMR = 0.010–0.070.

This variation is notable because all providers expose the same user-facing “deterministic” parameters (temperature zero, fixed seed where supported). The differences therefore reflect production serving infrastructure invisible to users, consistent with editorial calls for explicit model version tracking and prompt disclosure[7]. Structured JSON extraction constrains the output space more tightly than free-text summarization, leaving fewer viable token sequences and less opportunity for divergence. This task-dependence suggests that open-ended generation tasks (creative writing, hypothesis formulation) will likely exhibit even larger variation.

Perplexity Sonar’s particularly low reproducibility reflects its search-augmented architecture, where real-time web retrieval introduces an additional source of variation that compounds model-internal non-determinism. This observation is particularly relevant as retrieval-augmented approaches become standard in scientific applications: the combination of non-deterministic generation with variable retrieval creates a multiplicative reproducibility challenge that researchers must account for.

Cloud deployment does not preclude reproducibility

A natural hypothesis is that cloud deployment itself (network latency, load balancing, shared hardware) causes the non-determinism. To test this, we designed a quasi-isolation probe: we evaluated the same LLaMA 3 8B architecture served via Together AI’s cloud endpoint (INT4 quantisation) under identical prompts, seeds, and temperature as the local deployment. If cloud hosting were the driver, this deployment should exhibit API-like non-determinism.

It does not. The cloud-served LLaMA 3 achieves EMR = 1.000 [1.00, 1.00] for extraction and 0.880 [0.70, 1.00] for summarization, nearly identical to the local deployment on the same 10-abstract subset (EMR = 1.000 and 0.920, respectively; 95% bootstrap confidence intervals overlap). This provides evidence that cloud deployment *per se* does not cause non-determinism. The variability observed in GPT-4, Claude, and Gemini is instead consistent with the complexity of their production serving infrastructure.

Six well-documented mechanisms in distributed GPU inference can independently produce non-deterministic outputs even under greedy decoding (see Methods): non-associative floating-point arithmetic[15], mixed-precision accumulation

in BF16/FP16[11], multi-GPU tensor parallelism[8], FlashAttention kernel non-determinism[16], dynamic batching with continuous request scheduling[10], and speculative decoding[9]. Our single-GPU local deployment eliminates mechanisms 3–6, and GGML Q4 integer arithmetic mitigates mechanism 2, explaining near-perfect local reproducibility as a predicted consequence of the simpler execution environment. The Together AI result confirms that this is not an inherent limitation of cloud deployment: a well-controlled cloud serving environment can achieve near-local reproducibility. Deterministic execution modes are technically feasible, though they may entail performance trade-offs that current serving architectures prioritize differently.

Complex workflows amplify the reproducibility gap

Modern LLM applications rarely consist of single API calls. Multi-turn refinement dialogues and retrieval-augmented generation (RAG) pipelines are increasingly common in research workflows. To assess whether non-determinism compounds in these settings, we evaluated five models on a three-turn refinement task (extract, receive feedback, refine) and a RAG extraction task (extract structured fields from an abstract with a prepended retrieved context passage).

Local models maintain high reproducibility even under these more complex regimes (Fig. 2). Gemma 2 9B and Mistral 7B achieve perfect $\text{EMR} = 1.000$ for both multi-turn and RAG. LLaMA 3 8B shows $\text{EMR} = 0.880$ [0.76, 1.00] for multi-turn and 0.960 [0.88, 1.00] for RAG, slightly lower than single-turn, consistent with error propagation across dialogue turns where each response conditions the next.

Both API models exhibit near-zero reproducibility. Claude Sonnet 4.5 achieves $\text{EMR} = 0.040$ [0.00, 0.08] for multi-turn and 0.000 [0.00, 0.00] for RAG. Across 50 runs, not a single pair of RAG outputs was character-identical, with a mean normalized edit distance (NED) of 0.256. Gemini 2.5 Pro, despite supporting a seed parameter, achieves $\text{EMR} = 0.010$ [0.00, 0.03] for multi-turn and 0.070 [0.02, 0.13] for RAG. The convergence of two independent API providers, using different model architectures and serving stacks, on near-zero multi-turn reproducibility indicates that this is a general property of cloud-served inference for complex interaction regimes. This is concerning for agentic AI workflows, where LLMs are chained across multiple steps. If each step introduces independent non-deterministic variation, end-to-end reproducibility becomes unattainable without explicit provenance tracking at every node.

Outputs diverge textually but not semantically

If API outputs varied randomly, producing nonsensical or contradictory results, the problem would be easy to detect. Instead, our multi-level metric analysis reveals a subtler pattern (Fig. 3). Across all models and conditions, BERTScore F1 remains above 0.97 even when EMR approaches zero. ROUGE-L scores similarly remain high (>0.85 for all models), confirming substantial token-level overlap. This three-level dissociation (low bitwise identity, moderate surface divergence, high semantic preservation) defines “hidden” non-determinism: same meaning, different words. A researcher reading two outputs from the same prompt would judge them equivalent; only systematic comparison reveals that they are textually distinct.

Yet this hidden variation has practical consequences that extend beyond surface-level differences. A field-level divergence analysis of GPT-4 structured extraction outputs reveals that 100% of non-identical output pairs (24 of 30 abstract groups) differ in at least one conclusion-relevant field: objective, method, or key result (Extended Data Table 7). The `key_result` field diverges in 67% of groups, and `method` in 57%, meaning the textual variation is not limited to formatting or filler words but affects the substantive content that researchers would use for downstream analysis. For automated evidence synthesis pipelines, where outputs are parsed programmatically rather than read by humans, even minor lexical differences can propagate to different extracted values, different aggregated statistics, and ultimately different conclusions.

The implications extend beyond individual studies. Recent work has shown that LLMs used as evaluators of empathic communication are sensitive to subtle input variations[13], and psychometric assessments of LLM behavior reveal consistency patterns that depend on model architecture and fine-tuning approach[14]. Non-determinism is therefore not a technical curiosity but a property that interacts with how LLMs are deployed as measurement instruments, one that current evaluation frameworks do not account for. For the growing body of work that uses LLMs as annotators, evaluators, or data extractors, our results indicate that the “instrument” itself introduces measurement noise that is uncharacterized and unreported in standard methodology sections.

Temperature paradox: greedy decoding is not greedy for APIs

Temperature is the primary user-controllable parameter governing output variability. For local models, it behaves as theory predicts: under the temperature sweep (Fig. 4), local models show a clean monotonic decline from near-perfect EMR at $t = 0$ to $\text{EMR} \approx 0$ at $t = 0.7$. This confirms that local greedy decoding is truly greedy, collapsing the output distribution to a single deterministic sequence. API models, however, start from an already-low baseline at $t = 0$ and show a more complex pattern that violates the expected monotonicity.

Claude Sonnet 4.5 exhibits a non-monotonic response: extraction EMR *increases* from 0.067 at $t = 0.0$ to 0.700 at $t = 0.3$ before declining to 0.133 at $t = 0.7$. This counterintuitive result, where adding randomness *improves* reproducibility, suggests that Anthropic’s $t = 0$ decoding path exposes more infrastructure-level stochasticity than a small positive temperature that activates a more stable sampling pathway. GPT-4 shows a less dramatic but qualitatively similar pattern: the difference between $t = 0$ and $t = 0.3$ is smaller for API models than for local models, because the API baseline is already far from deterministic. At $t = 0.7$, local and API models converge toward uniformly low EMR, but through different trajectories: local models decline monotonically from near-1.0, while API models follow a flatter curve from an already-degraded baseline.

This temperature paradox has important practical implications. Researchers who set temperature to zero expecting deterministic outputs are, for API models, not achieving the greedy decoding they intend. The temperature–reproducibility relationship for API models depends on provider-specific implementation details that are opaque to users and undocumented in API references[12], making it impossible for

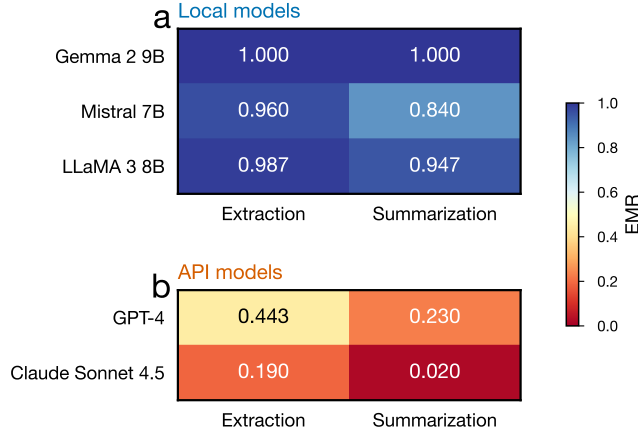


Fig. 1 Exact match rates under greedy decoding reveal a four-fold local-API gap. Heatmap of EMR for eight model deployments across extraction and summarization tasks under temperature = 0. Local models (top; blue) achieve EMR ≥ 0.840 , with Gemma 2 9B at a perfect 1.000. API-served models (bottom; red) range from 0.800 (DeepSeek) to 0.010 (Perplexity). All values are computed under each model’s representative greedy condition (C1 for local models and Claude; C2 for GPT-4; see Methods).

Table 1 Exact match rate under greedy decoding with 95% bootstrap confidence intervals. For local models, values reflect the fixed-seed condition (C1); for GPT-4, the variable-seed greedy condition (C2); for Claude, C1 (no seed parameter supported). Bootstrap: 10,000 resamples, percentile method. n = number of abstracts per group.

Model	Deployment	n	Extraction EMR	Summarization EMR
Gemma 2 9B	Local	10	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]
LLaMA 3 8B	Local	30	0.987 [0.96, 1.00]	0.947 [0.89, 0.99]
Mistral 7B	Local	10	0.960 [0.88, 1.00]	0.840 [0.72, 0.96]
DeepSeek Chat	API	10	0.800	0.760
GPT-4	API	30	0.443 [0.32, 0.57]	0.230 [0.16, 0.30]
Claude Sonnet 4.5	API	10	0.190 [0.05, 0.40]	0.020 [0.00, 0.05]
Perplexity Sonar	API	10	0.100	0.010
<i>Local average</i>			0.982	0.929
<i>API average</i>			0.383	0.255

researchers to predict or control the degree of non-determinism in their experiments without empirical measurement.

Discussion

Our findings reveal a pervasive and previously invisible reproducibility gap in LLM-based research. Under the settings that API documentation presents as deterministic (temperature zero, fixed seed), API-served models fail to reproduce their own outputs approximately four out of five times. This gap persists across five independent providers, extends to multi-turn and RAG workflows, and survives rigorous statistical correction. The immediate implication is that a substantial portion of published research relying on API-based LLMs may contain non-reproducible results without the

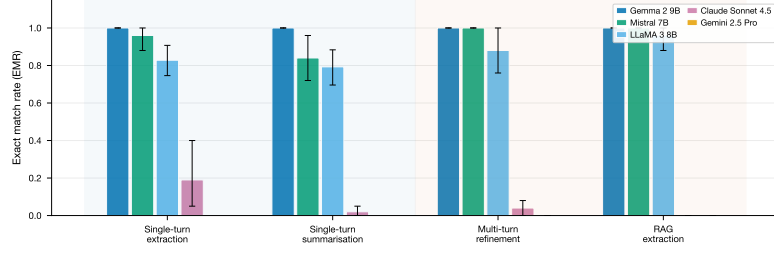


Fig. 2 Complex interaction regimes amplify the local-API reproducibility gap. EMR under greedy decoding ($C1, t = 0$) for five models across four scenarios: single-turn extraction, single-turn summarization, multi-turn refinement, and RAG extraction. Local models (Gemma 2, Mistral, LLaMA 3) maintain $\text{EMR} \geq 0.880$ across all scenarios. Both API models (Claude Sonnet 4.5 and Gemini 2.5 Pro) exhibit near-zero EMR, with Claude achieving 0.000 for RAG (50 runs, zero exact matches). Error bars: 95% bootstrap CIs.

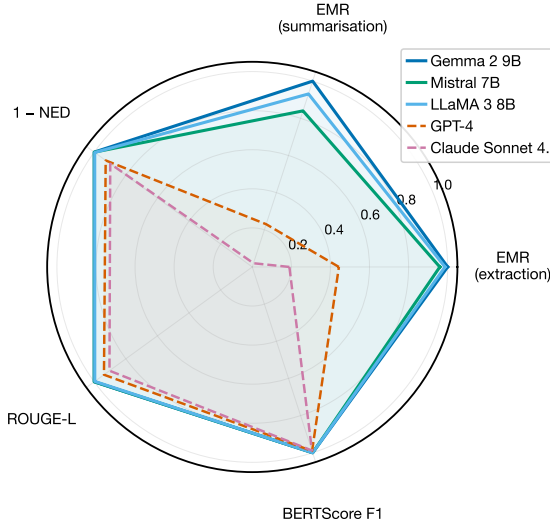


Fig. 3 API non-determinism is textual, not semantic. Three-level reproducibility profiles under greedy decoding. Local models (solid lines) occupy the outer region across all metrics. API models (dashed lines) show pronounced deficits in EMR and NED while maintaining BERTScore $F1 > 0.97$ (hidden non-determinism). Axes: EMR (extraction), EMR (summarization), 1-NED, ROUGE-L, BERTScore F1.

222 authors’ knowledge, a concern amplified by the growing use of LLMs in peer review[3],
 223 hypothesis generation[4], and agentic pipelines where non-determinism compounds at
 224 every step.

225 Prior work documented LLM non-determinism in isolated settings: inconsistent
 226 NLP benchmark outputs[27], failure of “deterministic” API settings[28], and non-
 227 deterministic code generation[29]. Our study advances beyond these observations
 228 by systematically comparing local versus API deployments across five providers,
 229 extending measurements to multi-turn and RAG workflows, and providing a causal

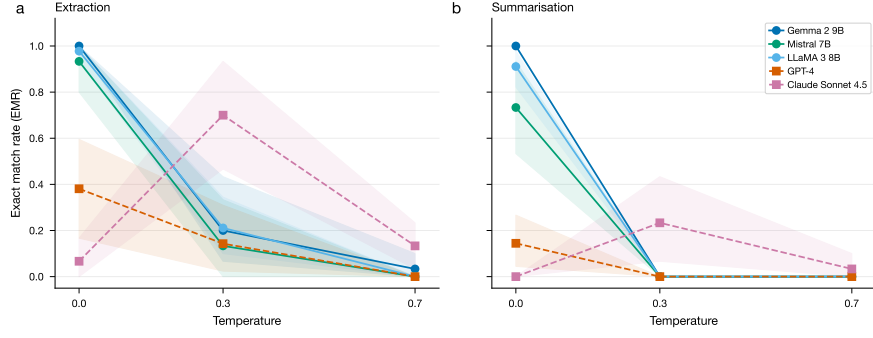


Fig. 4 Temperature paradox: greedy decoding is not greedy for API models. EMR versus temperature for five models (three local, two API). **a**, Extraction. **b**, Summarization. Local models (solid lines) show the expected monotonic decline from near-perfect EMR at $t = 0$. API models (dashed lines) start from an already-low baseline. Claude Sonnet 4.5 (red dashed) exhibits a non-monotonic anomaly: extraction EMR *increases* from 0.067 at $t = 0$ to 0.700 at $t = 0.3$.

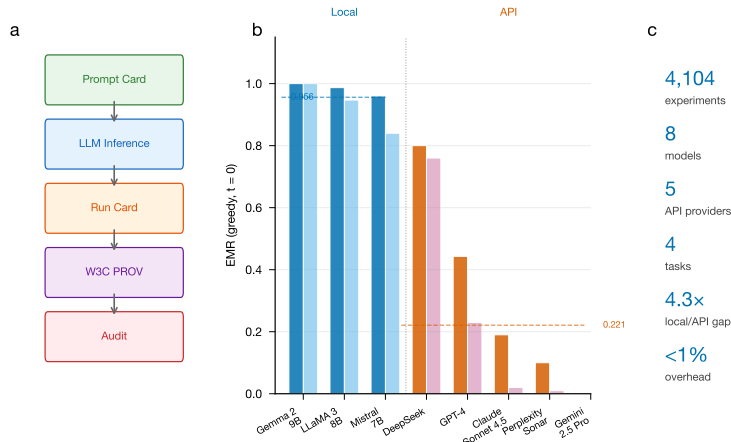


Fig. 5 Study overview and provenance protocol. **Left:** The provenance pipeline, from Prompt Card creation through Run Card logging and W3C PROV graph generation. **Centre:** EMR under greedy decoding for eight model deployments, illustrating the local-API reproducibility gap (local models green, $\text{EMR} \geq 0.840$; API models red, $\text{EMR} \leq 0.800$). **Right:** Key statistics: 4,104 experiments, 9 deployments, 7 execution environments, 4 tasks, <1% protocol overhead.

230 attribution framework that distinguishes infrastructure complexity from cloud deploy-
 231 ment.

232 The Together AI quasi-isolation result provides a key mechanistic insight: because
 233 the same model architecture achieves near-local reproducibility via a different cloud
 234 API, the non-determinism observed in GPT-4, Claude, and Gemini is attributable to
 235 production serving complexity rather than cloud deployment itself. This suggests that
 236 reproducibility is a design choice that providers could address through deterministic
 237 execution modes or transparent documentation, rather than an inherent limitation.

Documenting temperature zero as “deterministic” without disclosing infrastructure-level variation creates a false sense of reproducibility that may be more damaging than openly acknowledged stochasticity.

These findings also have regulatory implications. The EU AI Act[47] requires traceability for high-risk AI systems, and the NIST AI RMF[48] emphasizes transparency; hidden non-determinism directly undermines both. Ethical guidelines for LLM use in academic writing[20] similarly assume that documented configurations produce predictable outputs, an assumption our results show to be unwarranted.

Although the semantic preservation we observe (BERTScore F1 > 0.97) suggests that output quality is not corrupted, our field-level analysis shows that even “semantically equivalent” outputs differ in conclusion-relevant fields; the 78% non-match rate under “deterministic” settings represents a serious obstacle for regulatory submissions, systematic reviews, and automated pipelines.

The provenance protocol we provide offers a practical path forward. Grounded in W3C PROV[6] and adding less than 1% overhead (Extended Data Table 6), it creates auditable records linking every output to its generation context. Its key property is *differential diagnosis*: when all hashes match except the output hash, the divergence is automatically attributed to the generation process, a result confirmed across all 4,104 runs (Supplementary Section S4). This implements what recent editorials have advocated[7] and extends Model Cards[17] and Datasheets[18] to the inference layer. We propose a minimum reporting standard (model identity, parameters, reproducibility metrics, deployment mode, and output hashes) that our protocol automates at negligible cost (Supplementary Information).

Our study has several limitations. It covers eight models across four tasks but excludes code generation, mathematical reasoning, and creative writing. Input data comprises 30 English-language AI/ML abstracts; other languages and domains may show amplified effects. GPT-4 experiments used the `gpt-4-0613` snapshot; newer versions may differ, though the infrastructure-level mechanisms we identify are general. Multi-turn and RAG experiments include only two API providers; extending to others would strengthen generality. Sample sizes (10–30 abstracts per model) are well-powered for the large observed effects ($d > 1.6$) but may miss subtler phenomena.

Indeed, preliminary evidence from an applied domain suggests that the effects reported here have direct consequences for scientific conclusions. In a companion study applying six LLMs to a 500-abstract corpus of epidemiological studies on fine particulate matter (PM_{2.5}) and respiratory health[51], we find that extraction EMR under identical deterministic settings drops to 0.050 (Claude), 0.150 (GPT-4.1), and 0.200 (Gemini 2.5 Pro)—consistent with, and in some cases worse than, the values reported here. Critically, a meta-analytic propagation analysis reveals that up to 23 study-level effect estimates—each representing a distinct epidemiological finding on air pollution and respiratory outcomes—appear or disappear from the evidence base depending solely on which run of the LLM is used. This confirms that the hidden non-determinism documented in the present study is not confined to AI/ML benchmarks but propagates into applied health research where it can alter the composition of evidence available for clinical guidelines and regulatory decisions.

In summary, the “deterministic” settings offered by major LLM API providers do not deliver determinism in practice, creating a hidden reproducibility gap that spans five providers, persists across multi-turn and RAG workflows, and remains invisible because semantic equivalence masks textual divergence. Our quasi-isolation experiment attributes the cause to production infrastructure complexity rather than cloud deployment itself. The lightweight provenance protocol we provide, grounded in W3C PROV and requiring less than 1% overhead, makes this variation visible, auditable, and attributable, offering an actionable path toward reproducible LLM-based research. As large language models become embedded in clinical decision support, environmental monitoring, engineering design, and automated data analysis, the inability to reproduce a model’s output is not merely an academic inconvenience but a threat to the trustworthiness of systems that depend on it. Without explicit provenance tracking, the reliability of any LLM-based pipeline built on API inference cannot be assumed; it must be measured, documented, and continuously monitored.

Methods

Protocol design

The provenance protocol addresses the question: what is the minimum metadata needed per generative AI run to enable reproducibility assessment, auditing, and provenance tracking? The FAIR principles[49] provide a foundation for data stewardship, yet no standard exists for documenting the full context of generative AI outputs. Existing experiment-tracking tools such as MLflow[50] were designed for training pipelines and numerical metrics, not for inference-time text-generation provenance. Data leakage analysis across 17 ML fields[25] underscores the urgency of structured documentation. It comprises four components.

Prompt Cards are versioned documentation artifacts capturing design rationale and metadata for prompt templates, including SHA-256 hash, version, task category, assumptions, limitations, and target models. The concept extends Model Cards[17] and Datasheets for Datasets[18] to the prompt layer.

Run Cards capture the complete execution context of a single generative AI run through 24 core fields organized into five groups: identification (run ID, prompt hash, prompt text), model context (name, version, weights hash), parameters (temperature, seed, decoding strategy, parameters hash), input/output (text and SHA-256 hashes), and execution metadata (environment fingerprint, timestamps, duration, logging overhead). For API models, optional extension fields capture provider-specific metadata: API request ID, response headers, resolved model version, and a `seed_status` field distinguishing between seeds that were “sent”, “logged-only” (recorded for protocol parity but not transmitted, as with Claude), or “not-supported”.

W3C PROV integration. Each experimental group is translated into a PROV-JSON document[6] expressing generation provenance as a directed graph of entities (Prompt, InputText, ModelVersion, InferenceParameters, Output), activities (Run-Generation), and agents (Researcher, SystemExecutor). The formal semantics of PROV enable automated traversal and comparison, identifying the exact factor that differs between non-identical outputs without custom parsing.

Reproducibility checklist. A 15-item checklist organized into four categories (Prompt Documentation, Model and Environment, Execution and Output, Provenance) enables self-assessment.

We formally define the protocol as a tuple $\mathcal{P} = (PC, RC, G, CL)$ and prove an *audit completeness* property: for 10 defined audit questions, every question is answerable if and only if all field groups are populated. An ablation analysis confirms *minimality*: removing any field group renders at least one question unanswerable (Extended Data Table 6).

Models and infrastructure

We evaluate nine model deployments across three paradigms.

Local models (Ollama v0.15.5, Apple M4, 24 GB unified memory, macOS 14.6, Python 3.14.3): LLaMA 3 8B[30] (Q4_0), Mistral 7B[31] (Q4_0), Gemma 2 9B[32] (Q4_0). Weights hashes recorded via the Ollama API.

API-served models: GPT-4 (gpt-4-0613)[33] via OpenAI API; Claude Sonnet 4.5 (claude-sonnet-4-5-20250929)[34] via Anthropic API (urllib, no SDK); Gemini 2.5 Pro (gemini-2.5-pro-preview-05-06)[35] via Google AI Studio REST API (urllib); DeepSeek Chat via OpenAI-compatible API; Perplexity Sonar via Perplexity API.

Quasi-isolation probe: LLaMA 3 8B via Together AI (INT4, same architecture as local, meta-llama/Llama-3-8b-chat-hf).

Tasks

Four tasks spanning the output-structure spectrum: **Task 1, Scientific summarization:** produce a three-sentence summary of a scientific abstract. **Task 2, Structured extraction:** extract five fields (objective, method, key_result, model_or_system, benchmark) as JSON. **Task 3, Multi-turn refinement:** three-turn dialogue (extract, receive feedback, refine). **Task 4, RAG extraction:** structured extraction with a prepended retrieved context passage.

Input data

Thirty widely cited AI/ML abstracts (including Transformer, BERT, GPT-3, T5, and Chain-of-Thought[36–40]), varying in length (74–227 words) and technical complexity.

Experimental conditions

Five conditions systematically vary reproducibility factors: **C1** (fixed seed, greedy): $t = 0$, seed = 42, 5 repetitions; **C2** (variable seeds, greedy): $t = 0$, seeds = {42, 123, 456, 789, 1024}, 5 repetitions; **C3** (temperature sweep): $t \in \{0.0, 0.3, 0.7\}$, 3 repetitions each. Tasks 1–2 under all conditions for five models with full coverage; Tasks 3–4 under C1 for local models, Claude, and Gemini. DeepSeek and Perplexity: C1 only on Tasks 1–2. Together AI: C1 and C2 on Tasks 1–2. Grand total: 4,104 logged runs across 9 deployments and 7 execution environments.

363 For API models, the seed parameter is advisory (OpenAI[12]), absent
364 (Anthropic), or empirically insufficient (Gemini); greedy decoding does not guarantee
365 determinism[29].

366 Metrics

367 **Exact Match Rate (EMR)**: fraction of all $\binom{n}{2}$ output pairs within a group that are
368 character-identical. **Normalized Edit Distance (NED)**: Levenshtein distance[42]
369 normalized by the longer string. **ROUGE-L F1**: longest common subsequence
370 overlap[41]. **BERTScore F1**: embedding-based semantic similarity[43]. For struc-
371 tured extraction, we additionally compute JSON validity rate, schema compliance, and
372 field-level accuracy. Protocol overhead: logging time (ms), storage (KB), and overhead
373 ratio (%).

374 Statistical analysis

375 All EMR values accompanied by 95% bootstrap confidence intervals (10,000 resamples,
376 percentile method, per-abstract EMR)[44]. Primary comparisons: Wilcoxon signed-
377 rank test (non-parametric) and paired *t*-test (parametric), with Holm–Bonferroni
378 correction across 68 hypothesis tests. Effect sizes: Cohen’s *d* (parametric) and Cliff’s
379 delta[45] (non-parametric). Power analysis confirms >0.95 for all primary comparisons
380 at the observed effect sizes ($d > 1.6$)[46]. Balanced 10-abstract subsample analysis
381 confirms robustness (local EMR = 0.953 vs. API EMR = 0.304, 3.1 $\times$).

382 Sources of non-determinism in distributed inference

383 Six well-documented mechanisms can independently produce non-deterministic out-
384 puts under greedy decoding in distributed GPU inference: (1) non-associative
385 floating-point arithmetic[15]; (2) mixed-precision accumulation in BF16/FP16[11];
386 (3) tensor parallelism and all-reduce non-determinism[8]; (4) FlashAttention kernel
387 non-determinism[16]; (5) dynamic batching and continuous request scheduling[10];
388 (6) speculative decoding[9]. Our single-GPU local deployment eliminates mechanisms
389 (3)–(6) and GGML Q4 integer arithmetic mitigates (2), explaining near-perfect local
390 reproducibility as a predicted consequence.

391 Protocol overhead

392 The protocol adds <1% overhead across all models profiled: mean logging time 21–
393 30 ms versus inference latency of 4–182 s. Storage: ~4 KB per run (16 MB total for
394 4,104 runs). The overhead is consistent across local and API deployment.

395 Reporting summary

396 Further information on research design is available in the Nature Portfolio Reporting
397 Summary linked to this article.

398 **Data availability.** All 4,104 run records (JSON), PROV-JSON provenance
399 documents, Run Cards, Prompt Cards, input data (30 abstracts with DOIs),

and generated figures are publicly available at <https://github.com/Roverlucas/genai-reproducibility-protocol> under CC-BY 4.0 license. Source data are provided with this paper.

Code availability. The reference implementation of the provenance protocol, all analysis scripts, and figure-generation code are publicly available at <https://github.com/Roverlucas/genai-reproducibility-protocol> under MIT License.

Supplementary information. The Supplementary Information (single PDF) contains: (S1) Full prompts for all four tasks; (S2) All 30 abstracts with DOIs; (S3) Retrieved contexts for RAG experiments; (S4) API payload documentation for all nine deployments; (S5) Protocol comparison table (our protocol vs. MLflow, W&B, DVC, OpenAI Evals, LangSmith); (S6) Ablation study: protocol minimality verification (10 audit questions $\times$ 8 field groups); (S7) Chat-format control experiment (200 runs); (S8) 15-item reproducibility checklist; (S9) Statistical test results (full Holm–Bonferroni table, 68 tests); (S10) Environment and provenance transparency documentation.

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- **Author contributions:** L.R. conceived the study, designed the protocol, developed the software, conducted all experiments and analyses, and wrote the manuscript. H.V.S. reviewed the manuscript and co-supervised the research. E.T.B. contributed to experimental design and data analysis. A.T.d.A. contributed to methodology design and statistical analysis. Y.d.S.T. supervised the research, contributed to methodology design, and reviewed the manuscript.
- **Competing interests:** The authors declare no competing interests.
- **Correspondence:** Correspondence and requests for materials should be addressed to Lucas Rover (lucasrover@alunos.utfpr.edu.br).
- **Data availability:** All 4,104 experimental records, provenance metadata, and input abstracts are available to reviewers during peer review via the project repository at <https://github.com/Roverlucas/genai-reproducibility-protocol>. Upon publication, an archived snapshot with a persistent DOI will be deposited in Zenodo under a CC-BY 4.0 licence. Source data for all figures and tables are included in the repository.
- **Code availability:** The full source code for the provenance protocol, experiment runners, analysis scripts, and figure generation is available to reviewers during peer review at <https://github.com/Roverlucas/genai-reproducibility-protocol>. Upon publication, the repository will be made publicly available under an MIT licence and an archived release with a persistent DOI will be deposited in Zenodo. A README file with installation and execution instructions is provided.

Extended Data

Extended Data Table 1. Full three-level reproducibility assessment (EMR, NED, ROUGE-L, BERTScore F1) for all models under greedy decoding.

Extended Data Table 2. API versus local summary statistics with Cliff’s delta effect sizes and bootstrap confidence intervals on the EMR ratio.

Extended Data Table 3. Temperature sweep: EMR at $t \in \{0.0, 0.3, 0.7\}$ for five models.

Extended Data Table 4. Balanced 10-abstract subsample robustness analysis.

Extended Data Table 5. Warm-up analysis: cold-start effect characterization.

Extended Data Table 6. Protocol overhead and ablation matrix (protocol minimality verification).

Extended Data Table 7. Conclusion divergence analysis: GPT-4 field-level differences.

Extended Data Fig. 1. Normalized Edit Distance comparison across all models and tasks.

Extended Data Fig. 2. Run Card schema and W3C PROV graph example for a single experimental group.

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